

Group No: 07

Final report ME4073

Atlas Axillia Co. (Pvt) Ltd

By

Index No.	Name	Marks from ()
210286R	KELAART D.J.	
210317P	KUMARASINGHE K.A.T.K.	
210327V	LAHIRU N.P.G.S.	
210377X	MANUKULASOORIYA C.A.	
210615D	SOMASIRI R.D.S.N.K.	

Lecturer/s	Prof. HKG Punchihewa & Dr. JR Gamage
Module code	ME4073
Module name	Industrial Engineering
Semester	Eight (8)
Intake	21

**Department of Mechanical Engineering
University of Moratuwa
Sri Lanka**

10th May 2026

Table of Contents

List of figures.....	4
List of Tables	4
1. Introduction.....	5
2. Background.....	6
3. Problem Definition.....	7
4. Aim and Objectives.....	8
4.1 Aim	8
4.2 Objectives	8
5. Methodology	9
6. Data Collection	10
6.1 Gel Pen Assembly Process Steps.....	10
6.2 Gel Pen Assembly Process Time Data.....	11
6.3 Front Cap Assembling Machine Production Data	11
6.4 Cap Machine Run Log Analysis	12
6.5 Refill Writing Process.....	13
6.6 Labour Wages	13
6.7 Current Layout.....	15
.....	15
7. Analysis of Data.....	16
7.1 Process Flow Analysis	16
7.2 Data Consistency and Cycle Time Conversion	16
7.3 Data Consistency and Cycle Time Conversion	16
7.4 Refill Writing Capacity Analysis.....	17
7.5 Refill Writing Capacity Analysis.....	18
7.6 Current Method Vs Improved Method	18
7.6 Assembly Line Efficiency.....	19
7.7 Plant simulation	20
8. Findings.....	24
9. Proposed Industrial Automation Solutions	26
9.1 Solution 1 — Automatic Rotary Sticker Applicator Machine.....	26
9.1.1 Problem Being Solved	26
9.1.2 Proposed Solution Description	26
9.1.3 Technical Specifications	27
9.1.4 Expected Performance Improvement.....	27
9.1.5 Cost–Benefit Analysis	28
9.2 Solution 2 — Semi-Automated Pen-to-Display-Card Loading Fixture	30

9.2.1 Problem Being Solved	30
9.2.2 Proposed Solution Description	30
9.2.3 Technical Specifications	30
9.2.4 Expected Performance Improvement.....	31
9.2.5 Cost–Benefit Analysis	32
9.3 Solution 3 — Improved Downtime Management for Cap and Assembling Machines	33
10. Discussion and Conclusion	34
10.1 Summary of Findings.....	34
10.2 Discussion of Proposed Solutions.....	34
10.2.1 Automatic Rotary Sticker Applicator	34
10.2.2 Semi-Automated Pen-to-Display-Card Loading Fixture	34
10.2.3 Downtime Reduction for Cap and Assembling Machines.....	35
10.3 Expected Budget	35
10.4 Limitations	35
11. Future Work.....	36
12. References.....	36

List of figures

Figure 1: Atlas Axillia Co. (Pvt) Ltd Logo.....	5
Figure 2: Atlas Products.....	5
Figure 3: INM Plant (Welisara Plant).....	6
Figure 4: Assembly process of gel pen.....	10
Figure 5: Bush Error (Camera Detection).....	12
Figure 6: Rejection Box.....	13
Figure 7: Current layout.....	15
Figure 8: Model structure in the plant simulation.....	21
Figure 9: Plant simulation.....	22
Figure 10: Number of outputs.....	24

List of Tables

Table 1:Data collected on gel pen assembly process.....	11
Table 2:Data collected on front cap	11
Table 3: Machine parameters.....	12
Table 4: Errors happening in process.....	12
Table 5: Refill process steps	13
Table 6: Labour wages.....	14
Table 7: Cycle times per process	16
Table 8: Main station setting in the simulation.....	21
Table 9: The recommended simulation scenarios.....	23
Table 10:Simulation findings of current way	24
Table 11: Technical data of the machine	27
Table 12: Expected Performance Improvements of the automatic machine	27
Table 13: Estimated cost with the solution	28
Table 14: Technical data of solution 02.....	31
Table 15:Performance measurement of the machine	31
Table 16: Cost analysis of the machine	32

1. Introduction

Atlas Axillia Company Limited is one of the leading manufacturing companies in Sri Lanka, well known for its high-quality stationery and innovative products. Established in 1959, the company initially focused on school supplies and has grown into a diversified manufacturing organization producing exercise books, pens, pencils, and office products widely used across the country.

The company operates key production facilities, including its manufacturing plant in Welisara, which mainly focuses on plastic injection moulding. This plant produces plastic components such as pen bodies and other stationery parts. These semi-finished products are then transported to the assembly plant in Peliyagoda, where they are assembled into final products and packaged for distribution.

Through this integrated manufacturing and assembly process, Atlas ensures efficient production, consistent quality, and timely delivery, maintaining its strong reputation as a trusted household brand in Sri Lanka.



Figure 1: Atlas Axillia Co. (Pvt) Ltd Logo

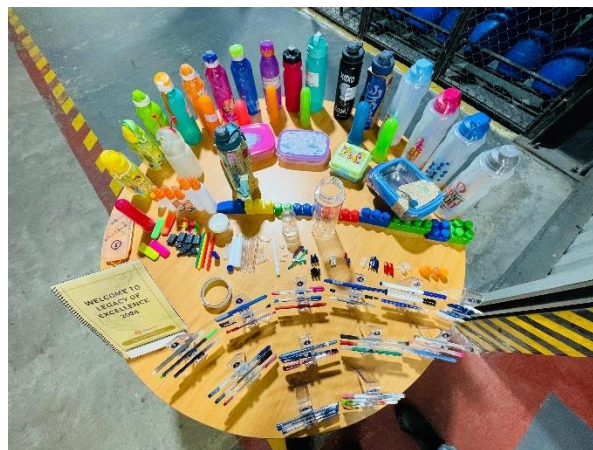


Figure 2: Atlas Products

2. Background

The assembly operations at the Atlas Axillia Company Limited plant located in Peliyagoda are a critical part of the plastic product cluster, primarily focusing on pen assembly and packaging. These processes are responsible for converting semi-finished components into final products suitable for market distribution.

The production process begins with receiving plastic components such as barrels, back plugs, nose cones, adapters, and caps from the Welisara manufacturing facility, where they are produced using injection moulding processes. After undergoing quality inspection, these components are released to the assembly lines.

In gel pen manufacturing, additional steps are required due to the higher viscosity of ink, including the use and removal of temporary ink stoppers and in-line writing tests for quality assurance. Following assembly, pens are transferred to packaging and display card attachment processes, where manual operations are still widely used. The integration of mechanical systems with manual labour allows operational flexibility, but also introduces variability in productivity and efficiency.



Figure 3: INM Plant (Welisara Plant)

3. Problem Definition

Within the plastic cluster, one of the key processes is pen assembly. The company manufactures a variety of pen models including Atlas Chooty, Atlas Gel Pen, Atlas Max, and Atlas Butter Gel. Although an assembly machine is available for the Atlas Gel pen, the process remains less productive than other pen assembly operations.

The gel pen assembly line suffers from two critical bottlenecks that significantly reduce its throughput and efficiency,

- Sticker Pasting

The sticker must be manually applied to each pen barrel. This is the slowest step on the gel pen line, achieving only 1,212 units/hour/head (2.97 seconds per unit), and sets the pace for the entire line.

- Attaching Pens to Display Cards (Card Packing)

Each display card holds thirty pens. Workers manually fit each pen into its slot on the card. Because a portion of the workforce consists of temporary employees with inconsistent skill levels, throughput is variable and often slow.

Additional contributing factors include,

- The front cap must be manually fitted to the barrel, slowing the overall cycle.
- A temporary ink stopper must be inserted and later removed during assembly due to the high viscosity of gel ink, adding complexity to the process.
- An in-line writing test is mandatory for quality assurance, interrupting the assembly flow.

The resulting line efficiency for the Gel Pen line stands at only 55.96%, with a balance delay of 44.04%. This significant imbalance means that faster workstations remain idle, waiting for the bottleneck stations to complete their operations, which directly reduces overall output and raises per-unit production costs.

4. Aim and Objectives

4.1 Aim

To optimize the gel pen assembly and display card packing processes at Atlas Axillia Lanka, Peliyagoda, by implementing industrial automation and process improvement solutions to eliminate identified bottlenecks, improve line balance, and increase overall production efficiency.

4.2 Objectives

1. To analyze and quantify the current inefficiencies in the sticker pasting and display card packing processes using time-study data and line-balance calculations.
2. To design and propose an automated sticker application system to replace manual sticker pasting, targeting a throughput improvement that eliminates the bottleneck.
3. To design and propose a semi-automated or automated display card loading mechanism to standardize output and reduce dependence on operator skill.
4. To evaluate the proposed automation solutions in terms of capital investment, payback period, and expected gains in throughput, line efficiency, and labor cost.
5. To recommend complementary process improvements including workstation re-balancing, operator redeployment, and quality-check integration to maximize the benefit of the proposed automation.

5. Methodology

The following methodology was adopted to systematically identify problems and develop engineering solutions,

- Factory visits and stakeholder discussions with the Production Manager, Assistant Production Manager, and Production Engineer were conducted to understand the current process and prioritize problem areas.
- Direct shop-floor time studies were used to measure output per operator per hour for each process step, enabling calculation of cycle time, line efficiency, and balance delay.
- Data from the gel pen production report including cap machine run logs, assembling machine run logs, and refill writing checklists were analysed to understand machine utilisation, downtime patterns, and error types.
- Industrial automation solutions were researched and evaluated for technical suitability, compatibility with existing equipment, and cost-effectiveness.
- Financial analysis (ROI, payback period) was performed for each proposed solution using current labor costs, production volumes, and estimated equipment costs.
- Recommendations were consolidated and presented with implementation priorities.

6. Data Collection

6.1 Gel Pen Assembly Process Steps

The complete assembly sequence for the Atlas Gel Pen is as follows,

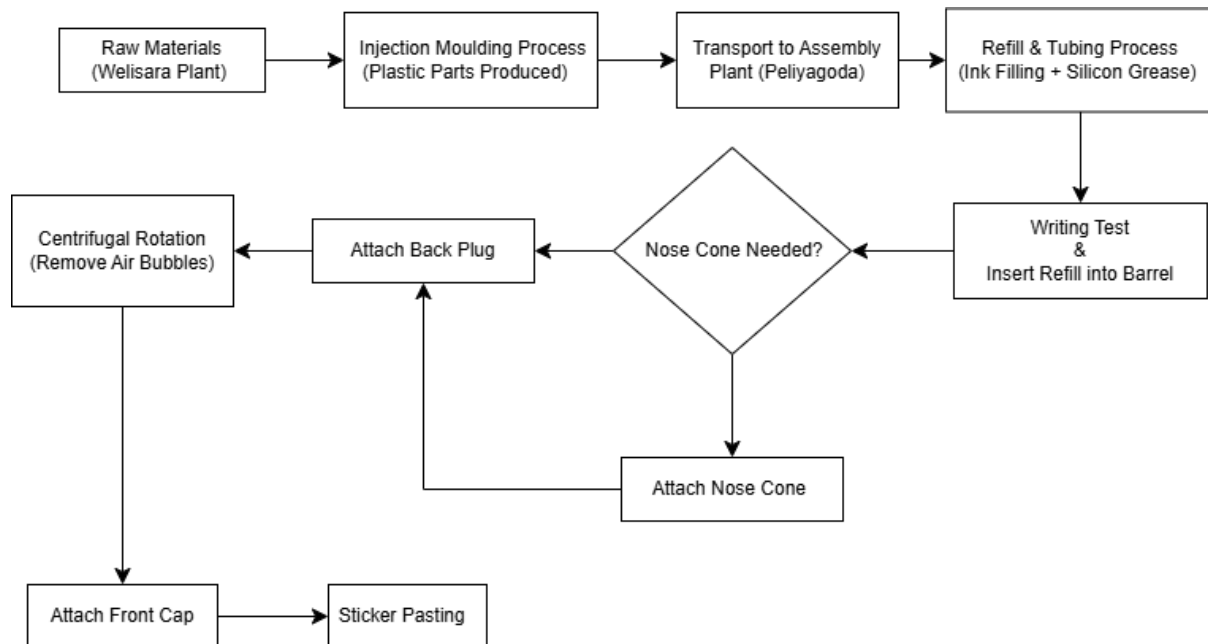


Figure 4: Assembly process of gel pen

1. Raw materials are available in Welisara, where the injection moulding plant is located. Plastic components of the pen such as barrel, back plug, cap, and nose cone are manufactured using injection moulding and stored in the Welisara warehouse.
2. These parts are then transported to the Peliyagoda assembly plant for assembly, refill, and tubing processes.
3. Ink is imported from Germany. The refill and tubing process is carried out, where ink is filled into the tube. Silicone grease is inserted from the back to trap the ink inside the tube. A writing test is then performed, and the completed refill is inserted into the barrel.
4. If the pen design includes a nose cone, it is attached.
5. The back plug is then attached to secure the refill inside the barrel.

6. The pen is subjected to a rotating (centrifugal) process to remove air bubbles. The ink moves toward the nose cone while air bubbles are pushed back through the silicone grease. (The ink cannot pass through the silicone grease.)
7. The front cap is attached.
8. Finally, the sticker pasting process is carried out.

6.2 Gel Pen Assembly Process Time Data

Time studies conducted on the gel pen assembly line machine yielded the following output rates.

Table 1: Data collected on gel pen assembly process

Pen Output From	Pen Output To	Time (min:sec:milisec)
700	770	0:37:68
800	874	0:40:11
874	1500	05:35:60
2150	2632	04:50:37

6.3 Front Cap Assembling Machine Production Data

Following data is extracted from the Front Cap Assembly Machine.

Table 2: Data collected on front cap

Cap Output From	Cap Output To	Time (min:sec:milisec)	Waiting Time for production line restart
50	120	00:33:44	0:05:22
120	267	1:37:22	0:04:47
300	464	1:47:27	1:30:80
464	512	0:38:05	0:39:83
516	1200	07:27:45	0:04:41
1231	1294	0:42:85	

Table 3: Machine parameters

Parameter	Value
Cap Machine Output Range	850 units per 12 minutes
Main Motor Speed	47 (normalized units)
Max Production Speed	92 caps per min
Assembling Machine Target	5,800 units in 90 minutes
Rejection Rate (Assembling)	44 rejected out of 5,800 (0.76%)

6.4 Cap Machine Run Log Analysis

Detailed run logs for the cap machine reveal frequent stoppages due to the following error types:

Table 4: Errors happening in process

Error Type	Description	Impact
Clip Error	Clip not aligned properly	Machine stoppage, manual correction required
Cap Error	Cap defective, machine fails to feed cap correctly	Rejected unit, line stoppage
Bush Error	Stopper position incorrect (detected by camera)	Automatic rejection, line stoppage
Output Error	Cap counted incorrectly (output count mismatch)	Inventory discrepancy, line stoppage

The run log shows significant waiting times. For example, one run segment recorded 1 minute 30 seconds of waiting due to a Bush Error. These stoppages reduce effective machine utilisation and compound the throughput problem at the bottleneck stations. Sometimes, the operator is not near by the machine so the time for error handling will be increased.



Figure 5: Bush Error (Camera Detection)



Figure 6: Rejection Box

6.5 Refill Writing Process

Table 5: Refill process steps

Step	Action
1	Check in assembled pen and assembled front caps
2	Remove the temporary stopper and check the writing
3	Attach the front cap and the barrel
4	Apply the sticker
5	Check out for pen handling for cardboard displaying boards

Time for checking in the pens for refill writing → 35 s

Time for checking the writing → 3500 pens/hr

Time for attaching the front caps → 1700 pens/hr

Display card packing (pen-to-card attachment) operates at approximately 900 units/hr which is even slower than sticker pasting. However, because this process runs in parallel with the main assembly line rather than in series, the gel pen line cycle time is formally set by sticker pasting. Both processes must be addressed to achieve a meaningful throughput improvement.

6.6 Labour Wages

Operators on the assembly line work in two shifts which are day shift (8:00 a.m. – 7:00 p.m.) and night shift (7:00 p.m. – 8:00 a.m.). The basic salary for entry-level operators is approximately Rs. 30,000 per month. Wage increases are tied to KPIs primarily covering attendance, individual output levels and Kaizen contributions.

Table 6: Labour wages

Category	Value
Basic Monthly Salary (Entry Level)	Rs. 30,000
Approximate Hourly Rate (based on 22 working days, 11 hrs)	Rs. 124
Shift Duration	11 hours active per shift

6.7 Current Layout

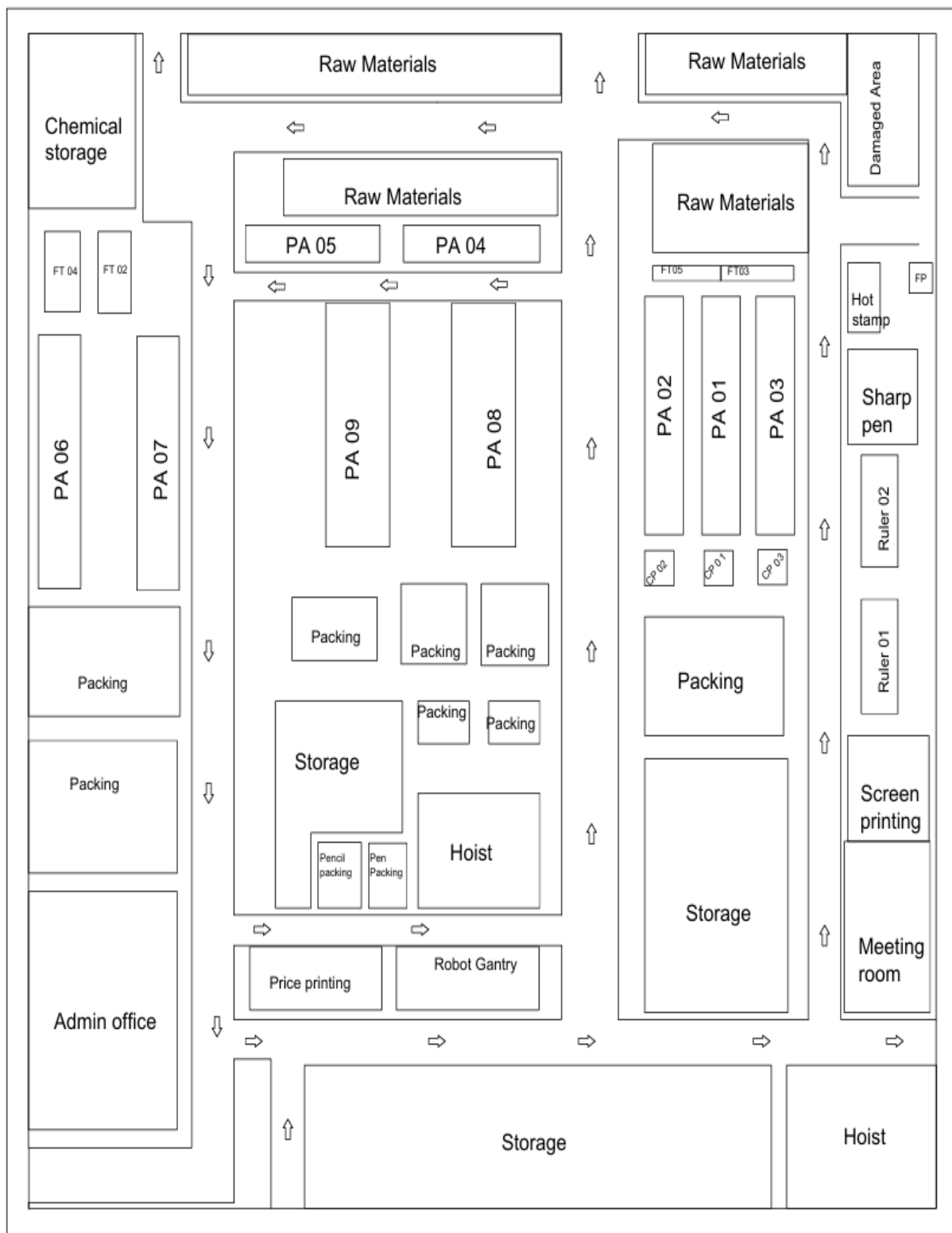


Figure 7: Current layout

7. Analysis of Data

7.1 Process Flow Analysis

The gel pen assembly process at Atlas Axillia consists of both machine-based and manual operations. The clip assembly process is considered a critical supporting process because the clip assembly speed is lower than the main pen assembly speed. Therefore, a buffer of clips is prepared before starting the pen assembly line. This buffer is used to prevent starvation of the downstream pen assembly process. In assembly line systems, inter-station buffers are commonly used to reduce the effect of starvation and blocking between workstations.

7.2 Data Consistency and Cycle Time Conversion

The collected production data were first converted into cycle time values. The cycle time per unit was calculated using,

$$\text{Cycle Time} = \frac{3600}{\text{Output per hour}}$$

Table 7: Cycle times per process

Process	Output Rate	Cycle Time
Refill writing / writing test	3500 pens/hour/person	1.03 s/pen
Clip insertion to pen	1700 pens/hour	2.12 s/pen
Sticker attachment	1700 pens/hour	2.12 s/pen
Clip insertion when sticker is attached before insertion	1800 pens/hour	2.00 s/pen

So, refill writing has a much lower cycle time than clip insertion and sticker attachment. Therefore, refill writing is not the bottleneck under the present operating condition.

7.3 Data Consistency and Cycle Time Conversion

At the start of production, the clip assembly machine is operated in advance to create a buffer of pen clips. For example, when 5000 pens are planned to be assembled, 8000 pen clips are prepared.

Parameter	Value
Planned pen quantity	5000 pens
Prepared clip quantity	8000 clips

Buffer quantity	3000 clips
-----------------	------------

$$\text{Buffer Quantity} = 8000 - 5000 = 3000 \text{ clips}$$

$$\text{Buffer Percentage} = \frac{3000}{5000} \times 100 = 60\%$$

Therefore, the clip preparation buffer is 60% higher than the immediate pen assembly requirement. This buffer is useful because it prevents the main pen assembly line from stopping when clip assembly is slower. At the current clip insertion rate of 1700 pens/hour, the extra 3000 clips provide approximately,

$$\text{Buffer Coverage Time} = \frac{3000}{1700} = 1.76 \text{ hours}$$

Therefore, the buffer can support the process for around 1 hour and 46 minutes beyond the immediate 5000 pen requirement.

7.4 Refill Writing Capacity Analysis

The refill writing or writing checking process has a capacity of 3500 pens/hour with one operator.

$$\text{Cycle Time} = \frac{3600}{3500} = 1.03 \text{ s/pen}$$

The transfer time from assembly to refill writing is 35 seconds. If this is considered as a batch transfer delay, it has a relatively small effect compared to the processing time of the full batch.

For a batch of 5000 pens:

$$\text{Refill Writing Time} = \frac{5000}{3500} = 1.43 \text{ hours}$$

$$1.43 \text{ hours} = 5148 \text{ seconds}$$

Total time including transfer:

$$5148 + 35 = 5183 \text{ seconds}$$

So, the transfer delay contributes only:

$$\frac{35}{5183} \times 100 = 0.68\%$$

Therefore, refill writing is not the main limitation of the system. The main focus should be on clip insertion and sticker attachment which have higher cycle times.

7.5 Refill Writing Capacity Analysis

According to assembly line balancing theory, the cycle time of a line is determined by the workstation with the longest processing time.

For the current gel pen process,

Process	Cycle Time	Bottleneck Status
Refill writing	1.03 s/pen	No
Clip insertion	2.12 s/pen	Possible bottleneck
Sticker attachment	2.12 s/pen	Possible bottleneck

The clip insertion and sticker attachment operations have the same cycle time of 2.12 s/pen. Therefore, these two operations form a bottleneck zone in the current process.

The maximum output rate of the bottleneck zone is,

$$\text{Line Capacity} = \frac{3600}{2.12} \approx 1700 \text{ pens/hour}$$

Therefore, even though refill writing can process 3500 pens/hour, the overall process flow after refill writing is limited to approximately 1700 pens/hour due to clip insertion and sticker attachment.

7.6 Current Method Vs Improved Method

Currently, the clip is inserted into the pen and then the sticker is attached. However, plant observations indicate that if the sticker is attached to the clip before clip insertion, the clip insertion rate increases from 1700 pens/hour to 1800 pens/hour.

Method	Clip Insertion Rate	Cycle Time
Current method	1700 pens/hour	2.12 s/pen
Sticker-before-insertion method	1800 pens/hour	2.00 s/pen

Improvement in output rate,

$$1800 - 1700 = 100 \text{ pens/hour}$$

Percentage improvement,

$$\frac{100}{1700} \times 100 = 5.88\%$$

Therefore, changing the method by attaching the sticker before clip insertion improves clip insertion productivity by approximately 5.88%.

For an 11-hour shift,

$$100 \times 11 = 1100 \text{ additional pens/shift}$$

For 22 working days per month,

$$1100 \times 22 = 24200 \text{ additional pens/month}$$

Therefore, if the improved method is applied continuously, it can increase production capacity by approximately 24200 pens per month per shift assuming the improved clip insertion stage becomes the controlling process.

Therefore, the method change is useful but to obtain the full benefit, sticker attachment should either be,

1. Carried out before final assembly as a separate buffered process
2. Supported by additional parallel operators or
3. Improved using a simple fixture or automation.

But also this may shift the bottleneck to the next slowest operation.

7.6 Assembly Line Efficiency

Considering the three main downstream operations after basic pen assembly:

Work Element	Cycle Time
Refill writing	1.03 s
Clip insertion	2.12 s
Sticker attachment	2.12 s

Total Work Content	5.27 s
--------------------	--------

The current line cycle time is governed by the slowest operation:

$$C = 2.12 \text{ s/pen}$$

Number of workstations considered:

$$N = 3$$

Line efficiency:

$$\begin{aligned} \text{Line Efficiency} &= \frac{\text{Total Work Content}}{N \times C} \times 100 \\ &= \frac{5.27}{3 \times 2.12} \times 100 \\ &= 82.86\% \end{aligned}$$

Balance delay:

$$\text{Balance Delay} = 100 - 82.86 = 17.14\%$$

This indicates that the selected process section is reasonably balanced between refill writing, clip insertion and sticker attachment. However, since clip insertion and sticker attachment are both slower than refill writing, they still control the maximum possible throughput.

7.7 Plant simulation

To further validate the time-study and line-balancing results, a discrete-event simulation model of the Atlas Gel Pen assembly process was developed using **Tecnomatix Plant Simulation**. The purpose of the simulation was to represent the real production flow, observe material movement between workstations, identify bottleneck behaviour, and evaluate the effect of buffer preparation before starting the pen assembly line.

The simulation model was developed based on the actual process sequence identified during data collection. The main flow begins with clip preparation and pen assembly as two separate input streams. The clip assembly process feeds a clip buffer, while the pen assembly process remains stopped until the required clip buffer quantity is available. After the buffer condition is satisfied, pens are released to the downstream process, where refill writing, clip insertion, sticker attachment, card packing, and finished goods collection are simulated. This logic was included because the report identified that a clip buffer is prepared before starting pen assembly to prevent starvation of the downstream process.

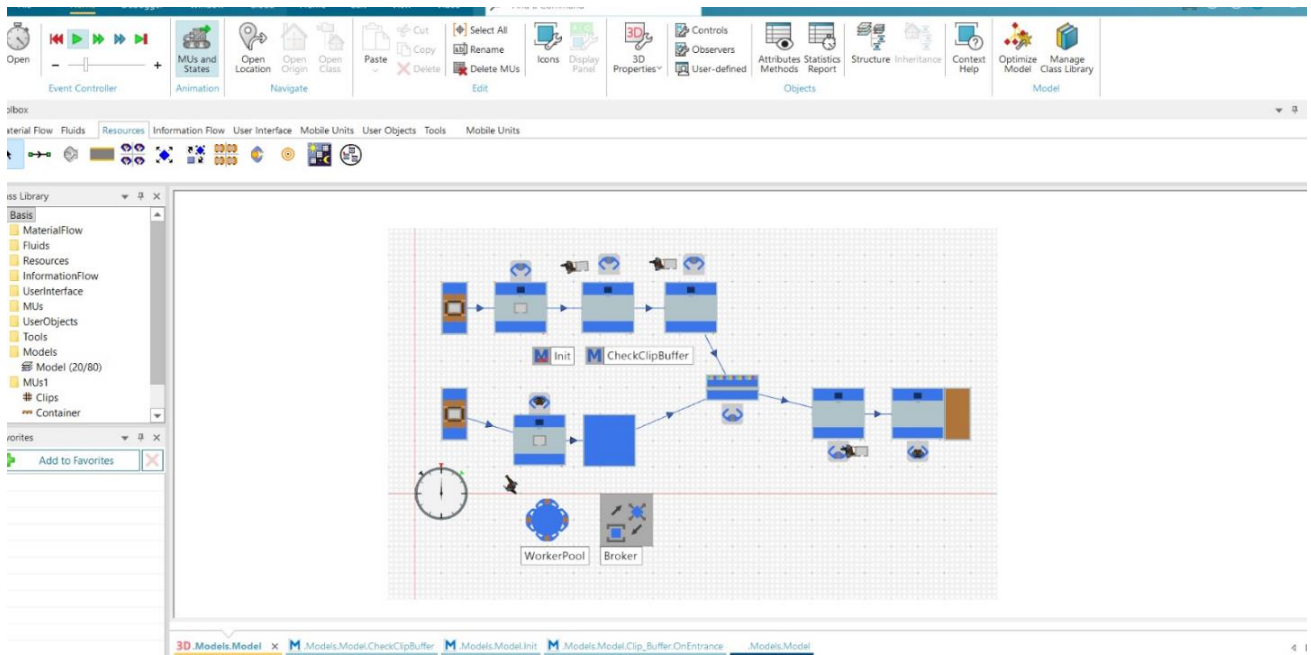


Figure 8: Model structure in the plant simulatione

The Clip_Buffer is used in the simulation to temporarily store completed clips. The Pen_Source is first stopped, and only released when the clip buffer achieves the desired starting stock level. This is the actual production process in which the clips are actually made prior to starting the main pen assembly process. In the simulation, the production of clips is kept up in the simulation while the pen line is operating, so that the buffer can provide for the gap in clip flow between clip demand and clip production.

Processing times for the model were determined from the measured production rates and converted to seconds per unit. The refill writing has 3500 pens/hour capacity, the clip insertion is about 1700 pens/hour, and the sticker attachment operation is one of the most bottleneck operations. Additionally, in the data collection section of the more detailed full-line model, the manual process time for major work elements such as front cap attachment, sticker pasting and display card packing are identified.

Table 8: Main station setting in the simulation

Process / Object	Plant Simulation Object	Input Data Used	Processing Time
Clip Assembly	SingleProc	Based on clip production requirement	3.60 s/clip for 1000 clips/hr case
Pen Assembly	SingleProc	Assembly machine data	0.56 s/pen
Transfer to Refill Writing	SingleProc / Transfer delay	Observed transfer/check-in time	35 s
Refill Writing	SingleProc	3500 pens/hr	1.03 s/pen
Clip Insertion	AssemblyStation	1700 pens/hr	2.12 s/pen

Process / Object	Plant Simulation Object	Input Data Used	Processing Time
Sticker Attachment	SingleProc	1212 units/hr/head or observed manual rate	2.97 s/pen
Card Packing	SingleProc	900 units/hr/head	4.00 s/pen
Finished Goods	Drain	Output counter	No processing time



Figure 9: Plant simulation

The Clip_Insertion station is modeled as an AssemblyStation since it needs two parts – one from the refill writing station and one from the clip buffer. The main movable part was defined as the pen and the clip was consumed during the assembly operation. Thus, the product of this station is a pen with the clip on it, and the clip then proceeds to the sticker attachment station.

Some significant performance metrics can be captured in the simulation. The most critical output is the amount of complete pens being placed in Finished_Goods. This value is the throughput of the production line simulated. Also, the station statistics include the working percent, waiting percent, blocked percent, and idle percent of each station. These results can be of use in checking whether the time-study analysis predicted a bottleneck, and if so, to see that it also occurs in the dynamic simulation.

The analysis of the line-balancing performed earlier is confirmed by the line-balancing behaviour observed in the simulation. The refill writing station is used after 22 high-speed clip insertion operations and refill sticker attaching operations, and is not expected to affect the line speed. The downstream operations, particularly pasting stickers and packing cards, take longer to complete, leading to waiting and work in progress accumulating up to those stations. This corroborates the

report's conclusion that the sticker pasting and packing of display cards need to be addressed to improve the throughput significantly.

Table 9: The recommended simulation scenarios

Scenario	Sticker Attachment Time	Card Packing Time	Purpose
Current process	2.97 s	4.00 s	Identify present bottleneck and WIP build-up
Sticker automation only	0.60 s	4.00 s	Check whether sticker bottleneck is removed
Card packing fixture only	2.97 s	1.00 s	Check whether packing output improves
Combined improvement	0.60 s	1.00 s	Evaluate total process improvement
Buffer optimization	Same as selected case	Same as selected case	Identify minimum clip buffer required

The Tecnomatix Plant Simulation model thus enables dynamic validation of the manual time-study analysis. It can serve as a way to illustrate how the clip buffer can increase the throughput for the pen assembly line, how the down stream operations of the bottle neck can be eliminated and how the proposed automation solutions can shift the bottle neck from sticker pasting and from packing cards to other down stream operations. This means the simulation can be used as a comparison tool of the current process with future process scenarios before any physical changes are made to the production floor.

8. Findings

Table 10: Simulation findings of current way

Simulation Indicator	Result Obtained	Interpretation
Simulation time	1 hour	Current process was evaluated for hourly output
Total output at Finished_Goods	1695 units	Approximate hourly production output
Throughput per hour	1695 units/hour	Close to theoretical 1700 units/hour capacity
Mean exit time	2.12 s	Output controlled by the 2.12 s bottleneck cycle time
Working percentage	87.58%	System was productive for most of the run
Waiting percentage	12.42%	Some waiting still exists in the line
Part type recorded	Clips	AssemblyStation main MU setting needs correction

To model the current gel pen assembly process before optimization, a model of the assembly was created using Technomatix Plant Simulation. The model encompassed the preparation of clips, assembly of pens, refill writing, clip insertion, sticker attachment, card packing and finished goods collection. The simulation was performed for 1 hour to come up with the current production output.

Object	Number of Entries	Number of Exits	Minimum Contents	Maximum Contents	Relative Empty	Relative Full	Relative Occupation without Interruptions	Relative Occupation with Interruptions
Finished_Goods	1695	1695	0	1	100.00%	-	0.00%	0.00%

Material Flow Properties

Object	Portion	Count	Sum	Mean Value	Standard Deviation
Finished_Goods	0.00%	1695	0.0000	0.0000	0.0000

Working Time

Object	Portion	Count	Sum	Mean Value	Standard Deviation
Finished_Goods	0.00%	0	0.0000	0.0000	0.0000

Set-up Time

Figure 10: Number of outputs

After 1 hour the Finished_Goods drain counted 1695 units. Therefore, the simulated throughput was,

$$\text{Throughput} = 1695 \text{ units/hour}$$

This result is very close to the theoretical output of the clip insertion process:

$$\frac{3600}{2.12} = 1698 \text{ units/hour}$$

Thus, the simulation shows that the present process can be completed at about 1700 units/hour, primarily because of the slowest part of the process, the clip insertion/sticker attachment bottleneck region.

The mean exit time was also plotted in the drain statistics being 2.12 s and this correlates with the clip insertion cycle time. This is a validation of the strong control of the line output by the 2.12 s bottleneck operation. The simulation also showed that with the synchronized stations and flow imbalance, some waiting still remains with working percentage of 87.58% and waiting percentage of 12.42%.

A limitation of the modelling was that the output part type for the Finished_Goods drain was detected as being clips rather than finished pens. This means that the AssemblyStation configuration should be corrected to make the pen the main MU and the clip the element to be consumed at the AssemblyStation. The throughput result is still useful however, as it is relatively comparable to the expected capacity of the bottleneck.

Overall, the simulation confirms the analytical result that the current gel pen line has limited capacity at approximately 1700 units/hour and efforts towards improvement should be directed towards the bottleneck operations in the downstream section of the line.

9. Proposed Industrial Automation Solutions

Two primary automation solutions are proposed to address bottleneck processes, followed by complementary improvements.

9.1 Solution 1 — Automatic Rotary Sticker Applicator Machine

9.1.1 Problem Being Solved

The sticker pasting process is performed entirely by hand. An operator picks a pre-cut sticker, aligns it with the pen barrel at the designated position, and presses it flat. At a rate of only 1,212 units/hour, this is the confirmed bottleneck for the entire gel pen assembly line.

9.1.2 Proposed Solution Description

Install a dedicated automatic rotary sticker applicator (label applicator machine) in line with the existing gel pen assembly conveyor. The machine would perform the following automatically:

Pen barrels are fed from the assembly conveyor into the applicator via a star-wheel or V-groove conveyor guide.

A photosensor detects the arrival of each pen barrel and triggers the label dispensing head.

A servo-driven label unwinding roll feeds the pre-cut sticker at the exact required pitch.

A vacuum pad or a wipe-on applicator applies the sticker to the barrel at the correct angular and axial position.

A pressure roller ensures full adhesion around the circumference of the barrel.

A vision system (basic 2D camera) performs an inline quality check to verify correct sticker placement and reject misaligned pens.

Accepted pens continue down the conveyor while the rejected pens are diverted to a separate bin.

9.1.3 Technical Specifications

Table 11: Technical data of the machine

Specification	Requirement
Machine Type	Rotary / Wrap-around Label Applicator
Target Throughput	6,000 – 9,000 units/hour (eliminating the bottleneck)
Label Stock	Compatible with existing pre-cut sticker rolls
Label Placement Accuracy	± 0.5 mm axial, $\pm 1^\circ$ rotational
Integration	In line with existing conveyor belt; 24V PLC controlled
Reject Mechanism	Pneumatic diverter gate, camera-triggered
Power Supply	Single-phase 220 V, ~2 kW
Footprint	Approx. 1.2 m $\times$ 0.5 m
Operators Required	1 (for sticker roll changeover and monitoring, down from 3)

9.1.4 Expected Performance Improvement

Table 12: Expected Performance Improvements of the automatic machine

Metric	Before (Manual)	After (Automated)
Sticker Pasting Throughput	1,212 units/hr	6,000–9,000 units/hr (machine)
Operators at Sticker Station	3 (to sustain line)	1 (supervisor/changeover)
Labour Saving (sticker station)	Rs. 60,000+/month	—
Sticker Placement Consistency	Variable (skill-dependent)	± 0.5 mm, consistent

Metric	Before (Manual)	After (Automated)
Rejection from Misapplied Stickers	Moderate	Near-zero (camera-verified)

9.1.5 Cost–Benefit Analysis

The following is an approximate cost-benefit analysis based on Sri Lanka market pricing for comparable label applicator equipment,

Table 13: Estimated cost with the solution

Item	Estimated Cost (Rs.)
Rotary label applicator machine (imported/local)	Rs. 800,000 – 1,500,000
Installation, conveyor integration & commissioning	Rs. 150,000 – 250,000
Vision inspection system	Rs. 200,000 – 400,000
Total Capital Investment (estimated)	Rs. 1,200,000 – 2,200,000

Savings Category	Monthly Saving (Rs.)
Labour saving - 2 operators redeployed (Rs. 60,000 each)	Rs. 120,000
Reduction in sticker wastage & rework	Rs. 10,000 – 20,000
Increased throughput revenue (conservative)	Rs. 30,000 – 50,000
Total Estimated Monthly Saving	Rs. 160,000 – 190,000

Payback Period Calculation Steps:

Step 1 — Total Monthly Savings (Conservative):

Labour saving (2 operators redeployed): $2 \times \text{Rs. } 60,000 = \text{Rs. } 120,000/\text{month}$

Sticker wastage & rework reduction: Rs. 10,000/month

Increased throughput revenue: Rs. 30,000/month

Total Monthly Saving (Conservative): Rs. 120,000 + Rs. 10,000 + Rs. 30,000 = Rs. 160,000/month

Step 2 — Total Monthly Savings (Upper Estimate):

Labour saving (2 operators redeployed): $2 \times \text{Rs. } 60,000 = \text{Rs. } 120,000/\text{month}$

Sticker wastage & rework reduction: Rs. 20,000/month

Increased throughput revenue: Rs. 50,000/month

Total Monthly Saving (Upper): Rs. 120,000 + Rs. 20,000 + Rs. 50,000 = Rs. 190,000/month

Step 3 — Payback Period (Conservative):

Payback Period = Capital Investment $\div$ Monthly Saving
= Rs. 1,200,000 $\div$ Rs. 160,000/month $\approx$ **7.5 months**

Step 4 — Payback Period (Upper Estimate):

Payback Period = Capital Investment $\div$ Monthly Saving
= Rs. 1,200,000 $\div$ Rs. 190,000/month $\approx$ **6.3 months** ($\approx$ 6–7 months)

Estimated Payback Period: Rs. 1,200,000 $\div$ Rs. 160,000/month $\approx$ 7.5 months (conservative scenario). Under the higher saving estimate, payback is achievable within 6–7 months.



Figure 1: Accutek APS-134 Automatic Horizontal Wrap Labeler — purpose-built for pen and cylindrical container labeling. Source: Accutek Packaging Equipment Inc. (accutekpackaging.com). Speed: up to 200 CPM (12,000 units/hr).

9.2 Solution 2 — Semi-Automated Pen-to-Display-Card Loading Fixture

9.2.1 Problem Being Solved

Attaching pens to display cards is a fully manual process. Each card holds 30 pens in specific clip slots. Workers push pens into the clips one by one. The throughput of approximately 900 units/hour/head is lower than that of any assembly machine step. Additionally, because a significant portion of card-packing workers are temporary employees with variable skill levels, both output rate and quality are inconsistent.

9.2.2 Proposed Solution Description

A purpose-built semi-automated pen loading fixture is proposed. The system operates as follows:

A purpose-built loading fixture holds one display card in a fixed, precise position. The card is loaded by the operator and locked in place with a quick-release clamp.

A set of 30 precisely machined guide channels, matching the spacing and angle of the card's pen clips, is mounted on a sliding carriage above the card.

Pens are loaded into the 30 guide channels simultaneously by the operator (or with the aid of a vibratory bowl feeder for bulk supply. Already implemented such vibrating and aligning mechanisms in the factory).

The operator activates a pneumatic press (foot pedal or push button). The press pushes all 30 pens downward through the guide channels into the card clips in a single stroke. (or small conveyor lines can be implemented but cost will be unnecessarily high)

A sensor array confirms all 30 pens are in place (presence detection). If any pen is missing, an alarm sounds and the card is not released.

The operator releases the clamp and removes the loaded display card.

This approach eliminates the need to push each pen in one at a time, reduces cycle time per card dramatically, and makes the process independent of individual operator dexterity.

9.2.3 Technical Specifications

Table 14: Technical data of solution 02

Specification	Requirement
Type	Semi-automatic pneumatic press with guide fixture
Cards per Hour Target	120–160 cards/hour (3,600–4,800 pens/hour per station)
Actuation	Pneumatic cylinder (available compressed air on factory floor)
Sensing	30× inductive or optical sensors for pen presence verification
Card Compatibility	Adjustable fixture to accommodate future card variants
Operators Required	1 per station (reduced from 2–3 for equivalent output)
Footprint	Approx. 0.8 m × 0.6 m benchtop unit
Power/Utilities	Compressed air (5–7 bar), 24V DC control circuit

9.2.4 Expected Performance Improvement

Table 15: Performance measurement of the machine

Metric	Before (Manual)	After (Semi-Automated Fixture)
Throughput	900 pens/hr	3,600–4,800 pens/hr/station
Cards per hour	30 cards/hr	60–80 cards/hr/station
Skill Dependency	High (variable with temp workers)	Low (fixture guides all operations)
Operators per station	2–3	1

9.2.5 Cost–Benefit Analysis

Table 16: Cost analysis of the machine

Item	Estimated Cost (Rs.)
Custom fixture fabrication	Rs. 200,000 – 400,000
Pneumatic cylinder, valves & air fittings	Rs. 50,000 – 80,000
Sensor array and control panel (PLC/relay)	Rs. 80,000 – 120,000
Installation and commissioning	Rs. 30,000 – 50,000
Total Capital Investment (estimated)	Rs. 360,000 – 650,000

Savings Category	Monthly Saving (Rs.)
Labour saving: 1–2 operators redeployed per station (Rs. 60,000 each)	Rs. 60,000 – 120,000
Reduction in rework from incorrectly loaded cards	Rs. 5,000 – 15,000
Increased throughput capacity	Rs. 15,000 – 30,000
Total Estimated Monthly Saving	Rs. 80,000 – 165,000

Payback Period Calculation Steps:

Step 1 — Total Monthly Savings (Conservative):

Labour saving (1 operator redeployed): $1 \times \text{Rs. } 60,000 = \text{Rs. } 60,000/\text{month}$

Rework reduction (incorrectly loaded cards): $\text{Rs. } 5,000/\text{month}$

Increased throughput capacity: $\text{Rs. } 15,000/\text{month}$

Total Monthly Saving (Conservative): $\text{Rs. } 60,000 + \text{Rs. } 5,000 + \text{Rs. } 15,000 = \text{Rs. } 80,000/\text{month}$

Step 2 — Total Monthly Savings (Upper Estimate):

Labour saving (2 operators redeployed): $2 \times \text{Rs. } 60,000 = \text{Rs. } 120,000/\text{month}$

Rework reduction (incorrectly loaded cards): $\text{Rs. } 15,000/\text{month}$

Increased throughput capacity: $\text{Rs. } 30,000/\text{month}$

Total Monthly Saving (Upper): Rs. 120,000 + Rs. 15,000 + Rs. 30,000 = Rs. 165,000/month

Step 3 — Payback Period (Conservative):

$$\begin{aligned}\text{Payback Period} &= \text{Capital Investment} \div \text{Monthly Saving} \\ &= \text{Rs. } 360,000 \div \text{Rs. } 80,000/\text{month} = \mathbf{4.5 \text{ months}}\end{aligned}$$

Step 4 — Payback Period (Upper Estimate):

$$\begin{aligned}\text{Payback Period} &= \text{Capital Investment} \div \text{Monthly Saving} \\ &= \text{Rs. } 360,000 \div \text{Rs. } 165,000/\text{month} \approx \mathbf{2.2 \text{ months}} \ (\approx 2\text{--}3 \text{ months})\end{aligned}$$

Estimated Payback Period: Rs. 360,000 ÷ Rs. 80,000/month ≈ 4.5 months (conservative). Under the higher saving estimate, payback is achievable within 2–3 months. This is the most cost-effective solution in the proposal.

9.3 Solution 3 — Improved Downtime Management for Cap and Assembling Machines

The cap machine run log reveals frequent and significant waiting times due to Clip Errors, Bush Errors, and Cap Errors. The following measures are recommended:

- **Preventive Maintenance Schedule:** Establish a formal weekly inspection of the clip alignment mechanism, cap feeder vibratory bowl, and camera-based bush detection system to catch wear before it causes line stoppages.
- **Spare Parts Buffer:** Maintain an on-site buffer of commonly failed parts (alignment clips, cap feeder springs, camera filters) to reduce stoppage-to-repair time.
- **Operator Error Log:** Digitize the error log so that error frequency trends can be tracked and used to guide maintenance priorities. Currently, errors appear to be logged manually on paper.

These low-cost interventions can reduce machine downtime drastically, directly improving effective throughput without additional capital investment.

10. Discussion and Conclusion

10.1 Summary of Findings

The study confirms that the Atlas Gel Pen assembly line at Peliyagoda operates at a line efficiency of only 55.96%, with the manual sticker pasting process identified as the primary bottleneck (1,212 units/hr/head; 2.97 s/unit). The display card packing process is a secondary constraint with even lower individual throughput (~900 units/hr/head) and high skill-dependency due to reliance on temporary workers.

Machine data from the production report reveals that the cap machine and assembling machine suffer from recurring errors (Clip, Bush, Cap, and Output errors) that cause significant unplanned waiting time, further compounding the throughput problem.

10.2 Discussion of Proposed Solutions

10.2.1 Automatic Rotary Sticker Applicator

This solution directly targets the confirmed bottleneck. A throughput of 6,000–9,000 units/hour from the machine versus the current 1,212 units/hour (and an effective 3,636 units/hour with 3 operators) represents a 65–148% increase in sticker station capacity. The station would no longer limit the line, shifting the bottleneck to front cap attachment. The inline vision check adds a quality control benefit not present in the current manual process. The estimated 6–7 month payback period is commercially attractive.

10.2.2 Semi-Automated Pen-to-Display-Card Loading Fixture

This solution offers the fastest payback (2–5 months) at the lowest capital cost (Rs. 360,000–650,000). Because it is a locally fabricated bespoke fixture rather than an imported machine, the company retains full flexibility to modify it as card designs evolve. The pneumatic press approach is robust, low-maintenance, and does not require specialized technicians to service. The sensor-verified completion feature eliminates the quality defect of partially loaded display cards, a defect that currently goes undetected until downstream or at the customer.

The key risk is the precision required in the guide channel fixture. If channel spacing or angle does not exactly match the display card clip geometry, pens will jam or clip improperly. Prototype testing with

the actual display card and multiple pen barrel sizes is essential before committing to full production of the fixture.

10.2.3 Downtime Reduction for Cap and Assembling Machines

The machine error log data suggests that a structured preventive maintenance and operator training programme can yield a considerable reduction in machine-related downtime at minimal cost. This is recommended as an immediate action independent of the capital investment solutions, as it will improve throughput and data quality within weeks of implementation.

10.3 Expected Budget

Solution	Estimated Capital Cost (Rs.)	Estimated Monthly Saving (Rs.)	Payback Period
Automatic Rotary Sticker Applicator	1,200,000 – 2,200,000	160,000 – 190,000	6–14 months
Semi-Auto Pen-to-Card Loading Fixture	360,000 – 650,000	80,000 – 165,000	2.5–8 months
Total	1,560,000 – 2,850,000	240,000 – 355,000	—

10.4 Limitations

- Time study data was collected over a limited observation period. Seasonal variations in operator performance and production mix may alter the baseline figures.
- Equipment cost estimates are based on market enquiries and published price ranges; actual quotes from suppliers may vary significantly.
- The payback calculations assume consistent production demand of approximately 500,000 units/month. Lower demand would extend the payback period.
- The Technomatrix simulation component of the project (layout analysis) is not covered in this report and will be addressed separately.

- Integration of the sticker applicator machine with the existing conveyor system requires detailed mechanical drawings of the current conveyor, which were not available at the time of report preparation.

11. Future Work

- Procure quotations from at least two local and one imported supplier for the rotary label applicator machine and evaluate total cost of ownership including spare parts availability in Sri Lanka.
- Commission a prototype of the pen-to-display-card loading fixture from a local engineering workshop and conduct at least 50-cycle acceptance testing before full deployment.
- Implement the preventive maintenance schedule for cap and assembling machines and collect error log data for 4 weeks to quantify downtime reduction.
- After automation of the sticker pasting station, conduct a repeat time study across all gel pen line stations to establish the new line balance and identify the updated bottleneck.
- Evaluate the feasibility of automating the front cap attachment step (currently 2.25 s/unit) as the next improvement phase, following the completion of the current project.
- Explore whether the same label applicator machine can be shared across other pen lines (Atlas Max, Atlas Butter Gel) to improve return on investment.

12. References

- [1] S. Prashan, Senior Executive Production, Atlas Axillia Co. (Pvt) Ltd, Peliyagoda, Sri Lanka. Personal communication, 2026.
- [2] M. P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing, 4th ed. Upper Saddle River, NJ, USA: Pearson Education, 2015.
- [3] B. W. Niebel and A. Freivalds, Methods, Standards and Work Design, 13th ed. New York, NY, USA: McGraw-Hill, 2014.
- [4] S. Shingo, A Study of the Toyota Production System from an Industrial Engineering Viewpoint. Portland, OR, USA: Productivity Press, 1989.

- [5] Accutek Packaging Equipment Company, Inc., “APS-134 Automatic Horizontal Wrap Labeler,” [accutekpackaging.com](https://www.accutekpackaging.com/labeling-machines/pressure-sensitive-labelers/aps-134/). [Online]. Available: <https://www.accutekpackaging.com/labeling-machines/pressure-sensitive-labelers/aps-134/>. [Accessed: May 2026].
- [6] Alibaba.com, “Round Pen Labeling Machine — Suppliers and Manufacturers,” [alibaba.com](https://www.alibaba.com/showroom/round-pen-labeling-machine.html). [Online]. Available: <https://www.alibaba.com/showroom/round-pen-labeling-machine.html>. [Accessed: May 2026].
- [7] Alibaba.com, “Ball Pen Labeling Machine — Suppliers and Manufacturers,” [alibaba.com](https://www.alibaba.com/showroom/ball-pen-labeling-machine.html). [Online]. Available: <https://www.alibaba.com/showroom/ball-pen-labeling-machine.html>. [Accessed: May 2026].
- [8] Janesville Tool and Manufacturing, Inc., “Pneumatic Presses — Product Category,” [janesvilletool.com](https://www.janesvilletool.com/C/5/PneumaticPresses). [Online]. Available: <https://www.janesvilletool.com/C/5/PneumaticPresses>. [Accessed: May 2026].
- [9] Janesville Tool and Manufacturing, Inc., “A-3066 Adjustable Precision Pneumatic Press — Technical Specifications,” [janesvilletool.com](https://www.janesvilletool.com/P/108/a-3066-adjustable-precision-pneumatic-press). [Online]. Available: <https://www.janesvilletool.com/P/108/a-3066-adjustable-precision-pneumatic-press>. [Accessed: May 2026].
- [10] Aaron Equipment Company, “Used Accutek APS-134 Horizontal Wrap Labeler — Equipment Listing,” [aaronequipment.com](https://www.aaronequipment.com/usedequipment/cannabis-equipment/packaging-equipment/accutek-packaging-equipment-aps-134). [Online]. Available: <https://www.aaronequipment.com/usedequipment/cannabis-equipment/packaging-equipment/accutek-packaging-equipment-aps-134>. [Accessed: May 2026].
- [11] Anchuang Machinery Technology Co., Ltd., “AC-320B Automatic Blister Paper Card Packing Machine,” [anchuangmachinery.com](https://www.anchuangmachinery.com/stationery-packing-machine-product/). [Online]. Available: <https://www.anchuangmachinery.com/stationery-packing-machine-product/>. [Accessed: May 2026].